

QIHAO (KEITH) LIN

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EDUCATION

Carnegie Mellon University (CMU), Pittsburgh, PA, USA
M.S. in Biomedical Engineering
Advisor: Prof. Zackory Erickson

Jan. 2026 – Present
GPA: 3.7/4.0

Sun Yat-sen University (SYSU), China
B.Eng. in Aeronautical and Astronautical Engineering

Sept. 2021 – Jun. 2025
GPA: 3.5/4.0

PUBLICATIONS AND MANUSCRIPTS

- [1] **Qihao Lin**, et al. “Uncover to Re-cover: Model-Based Blanket Manipulation for In-Bed Assistance.” Submitted to the *2027 ACM/IEEE International Conference on Human-Robot Interaction (HRI 2027)*.
- [2] **Qihao Lin**, Borui Chen, Yuping Zhou, Jianing Wu, Yulan Guo, Weishi Zheng, and Chongkun Xia. “Transparent Fragments Contour Estimation via Visual–Tactile Fusion for Autonomous Reassembly.” *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, under review. doi:10.48550/arXiv.2603.20290.
- [3] **Qihao Lin**, et al. “Standardized-Lexicon-Scale Chinese Sign Language Translation.” *Manuscript in preparation*, 2026.
- [4] Victor-Louis De Gusseme, Thomas Lips, . . . , **Qihao Lin**, . . . , Francis Wyffels. “A Dataset and Benchmark for Robotic Cloth Unfolding Grasp Selection: The ICRA 2024 Cloth Competition.” *The International Journal of Robotics Research (IJRR)*, 2026. doi:10.1177/02783649251414885.

RESEARCH EXPERIENCE

Robotic Caregiving and Human Interaction Lab, CMU
Graduate Researcher

Dec. 2025 – Present
Advisor: Prof. Zackory Erickson

Project: Uncover to Re-cover: Model-Based Blanket Manipulation for In-Bed Assistance

- Developed a model-based robotic system for sequential blanket uncovering and re-covering to support in-bed assistance.
- Introduced a recoverability-aware uncovering objective and fine-tuned graph-based cloth dynamics models on smoothing-focused interactions to improve downstream re-covering performance.
- Integrated multi-view RGB-D perception with a Hello Robot Stretch 3 and evaluated the system in a study with 10 participants.

Robot Technology & Intelligent Perception Group (RTIP), School of Advanced Manufacturing, Sun Yat-sen University
Undergraduate Researcher

Advisor: Prof. Chongkun Xia
Dec. 2023 – Sept. 2025

Project: Transparent Fragments Contour Estimation via Visual–Tactile Fusion for Autonomous Reassembly

Leader, Thesis

Sept. 2024 – Sept. 2025

- Developed a visual–tactile perception framework for estimating transparent-fragment contours and supporting precise matching and reassembly.
- Built an autonomous reassembly pipeline integrating perception, tactile reconstruction, and manipulation, and created TransFrag27K, a dataset for transparent-fragment grasping and reassembly.

AI&ROBOT LAB, Shenzhen International Graduate School, Tsinghua University

Project: Benchmarking Grasp Selection for Robotic Cloth Unfolding: The ICRA 2024 Cloth Competition

Key Member, Cloth Track of the 9th Robotic Grasping and Manipulation Competition

Feb. 2024 – Aug. 2024

- Developed and evaluated grasp-selection models for robotic cloth unfolding through model training, dataset preparation, and depth-image processing; contributed to an eighth-place competition result.
- Contributed to a U-Net-based grasp-selection approach and the resulting paper published in *The International Journal of Robotics Research (IJRR)*.

INDUSTRY EXPERIENCE

Shenzhen Haoya IoT Co., Ltd., Shenzhen, China
Machine Learning Engineer

haoyarl.com
Sept. 2025 – Present

- Developed recognition and online interaction components for a Chinese sign-language translation system covering approximately 8,000 entries in the national standard sign-language dictionary.
- Implemented motion-aware segmentation and real-time inference, with sentence-level translation via language-model-based word grouping.

HONORS AND AWARDS

Outstanding Undergraduate Thesis Award (University-Level), Sun Yat-sen University, 2025
Outstanding Undergraduate Thesis Award (School of Advanced Manufacturing), Sun Yat-sen University, 2025
Third-Class Academic Scholarship, Sun Yat-sen University, 2023–2024
Third-Class Academic Scholarship, Sun Yat-sen University, 2022–2023

TECHNICAL SKILLS

Programming & ML	Python, C/C++, MATLAB, PyTorch, graph neural networks
Robotics	ROS/ROS2, model-based control, simulation and sim-to-real, RGB-D perception, point-cloud processing, visual–tactile sensing